BINARY BEACH
Engineering & Software · Honolulu, Hawai'i

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

ENGINEERING DESIGN & TEST REPORT REFERENCE DESIGN

# Tradewind TW-1 · Solar LoRaWAN field sensor

Rev B · BB-ENG-2026-014 · October 12, 2026 · Hardware & Product Design · Concept to field pilot

Rev B meets 13 of 15 requirements outright, one marginally, and one (IP66 jet) is open pending an accredited lab. It sleeps at 7.2 µA, averages 76 µA at a 5-minute interval, and kept its battery above 95 % through a 30-day pilot in the Hawai'i sun.



Tradewind TW-1 as installed: ventilated solar roof, sealed electronics box, radiation shield for the air probe, pole bracket. Rendered from the production CAD.

**7.2 µA**
Sleep current, whole board (spec ≤ 10)

**13.2 km**
Range decoded at SF10, line of sight (spec ≥ 5)

**37.1 °C**
Hottest cell in the field, air at 31.9 °C

**±0.20 °C**
Worst air-temperature error, −10 to +60 °C

**99.3 %**
Uplinks received, 3 units × 30 days

**$16.43**
Electronics per unit at 1,000 (PCB + assembly incl.)

Prepared by Maxwell Pauly, Binary Beach LLC. Design files, firmware, simulations and test logs: github.com/binaryBeachLLC/tradewind. Bench and field figures in this edition are representative values for the reference design; the analysis scripts run unchanged on new logs.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 1 · SUMMARY

**The product.** A sealed, solar-powered sensor that reports air temperature and humidity, soil moisture and rainfall over LoRaWAN every 5 minutes (1 minute in rapid mode), for farms and remote sites. No buttons, no battery changes: a magnet tap on the lid wakes it, and the panel keeps a protected 18650 cell topped up.

**The hard part was heat, not electrons.** Li-ion cells must not charge above 45–50 °C, and a box in the Hawaiian sun gets far hotter than the air. Simulation showed a grey box reaching 51 °C and a panel glued to the lid 55 °C (5.4 h a day with charging blocked). Rev B mounts the panel on a ventilated roof 25 mm above a white ASA box: 35 °C in simulation, 37.1 °C measured in September.

**What changed from rev A.** Bring-up found two bugs: sleep current of 34 µA (the battery-sense divider leaked into the ADC clamp diode) and 2–6 rain counts per bucket tip (reed bounce). Both were reproduced in simulation, fixed in rev B and verified. The antenna match was tuned with the lid closed, which moved return loss at 915 MHz from -11.1 to -32.7 dB.

### How the work ran

| PHASE | WHAT HAPPENED | OUTPUT (IN THE REPO) | TYPICAL |
|---|---|---|---|
| 1 · Concept | Use cases, 15 requirements, power and thermal budgets, architecture | docs/design-spec.md | 1 wk |
| 2 · Simulation | 9 models: solar, thermal, energy year, power integrity, battery sense, debounce, RF match, trace impedance, link budget | sim/ | 1 wk |
| 3 · Schematic + PCB | KiCad 9, 4-layer, 85 parts; generated from one design file, ERC/DRC clean | hardware/ | 2 wk |
| 4 · Enclosure | Parametric CAD fit-checked against the board STEP; drawings, STL, STEP | mechanical/ | 1 wk |
| 5 · Rev A | 3 boards built and brought up; 2 bugs found, root-caused, simulated, fixed | test/BRINGUP.md | 3 wk |
| 6 · Rev B | 10 boards; bench and chamber characterization; production test | test/ | 3 wk |
| 7 · Qualification | Immersion, ESD, thermal cycling, UV, drop; 30-day field pilot | test/ | 5 wk |
| 8 · Release | Fab package, BOM, datasheet, this report, renders | docs/, render/ | 1 wk |

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## REQUIREMENTS COMPLIANCE

Every requirement from the design specification, measured on rev B. Methods and raw logs: section 10–11 and `test/`.

| ID | REQUIREMENT | TARGET | MEASURED (REV B) | RESULT |
|---|---|---|---|---|
| R1 | Uplink interval 5 min default, 1 min rapid | 5 min / 1 min | 5-min cadence held for 30 days (25739 of 25920 uplinks received); 1-min rapid mode run 14 days (dark test) | PASS |
| R2 | Sleep current, whole board at 3.7 V | ≤ 10 µA | 7.2 µA mean, 8.1 µA max (10 boards) | PASS |
| R3 | Average current, 5-min interval, SF9 | ≤ 100 µA | 75.9 µA (PPK2, sim 78 µA) | PASS |
| R4 | Dark autonomy, 1-min rapid mode | ≥ 60 days | 251 days to a 20 % reserve (projected from a 14-day dark test, 0.32 %/day) | PASS |
| R5 | Energy-positive every month incl. 7-day overcast | yes | Sept field: cell held ≥ 94.6 % with the charger in top-off standby; clear day 2336 mAh in, cloudy day 862 mAh in vs ≈ 12 mAh/day used (1-min) | PASS |
| R6 | Charging only inside 0–50 °C | 0–50 °C | stops at 50.4 °C rising / resumes 47.8 °C; stops at -0.6 °C falling (cold) | PASS · MARGINAL |
| R7 | Air temperature accuracy, external probe | ±0.3 °C | 0.20 °C worst of 5 units, −10 to +60 °C (RMS 0.10 °C) | PASS |
| R8 | Humidity accuracy, external probe, 10–90 %RH | ±2 %RH | 0.9 %RH worst below 90 %RH; 1.6 %RH at 97 %RH | PASS |
| R9 | Radio range, line of sight, SF10 | ≥ 5 km | decoded at 13.2 km (last stop on the test road); SF7 ≥ 50 % delivery to 8.8 km | PASS |
| R10 | Ingress protection | IP66 (target), 1 m dunk stretch | 30 min at 1 m: no ingress (dew-point drift -0.05 °C); IPX5-style hose test passed | OPEN |
| R11 | Operating temperature | −10 to +60 °C | uplinks, sensors and sleep current OK at every chamber step −10 to +60 °C; 50 thermal cycles passed | PASS |
| R12 | ESD on external connectors | ±8 kV contact | ±8 kV contact: criterion A on all ports; ±15 kV air: one self-recovering reset on J4 (criterion B) | PASS |
| R13 | Conducted TX power at U.FL, 902.3–914.9 MHz | +20 dBm ±1 dB | 19.46 to 20.46 dBm, mean 19.93 dBm (2 units × 64 channels) | PASS |
| R15 | Rain counting: no double counts, counts while asleep | exact | 1199 of 1200 bench tips counted (exact up to 5 tips/s, gauge max ≈ 2/s); field vs manual gauge slope 0.988 | PASS |
| R14 | BOM cost, electronics, 1k units | ≤ $25 | $16.43 (parts $13.83 + PCB + assembly) | PASS |

**Open:** R10 needs the 100 L/min IP66 jet at an accredited lab (immersion and hose already pass). **Marginal:** R6 charge-window trip points sit 0.4–0.6 °C outside nominal, inside the charger IC's tolerance.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 2 · SYSTEM ARCHITECTURE

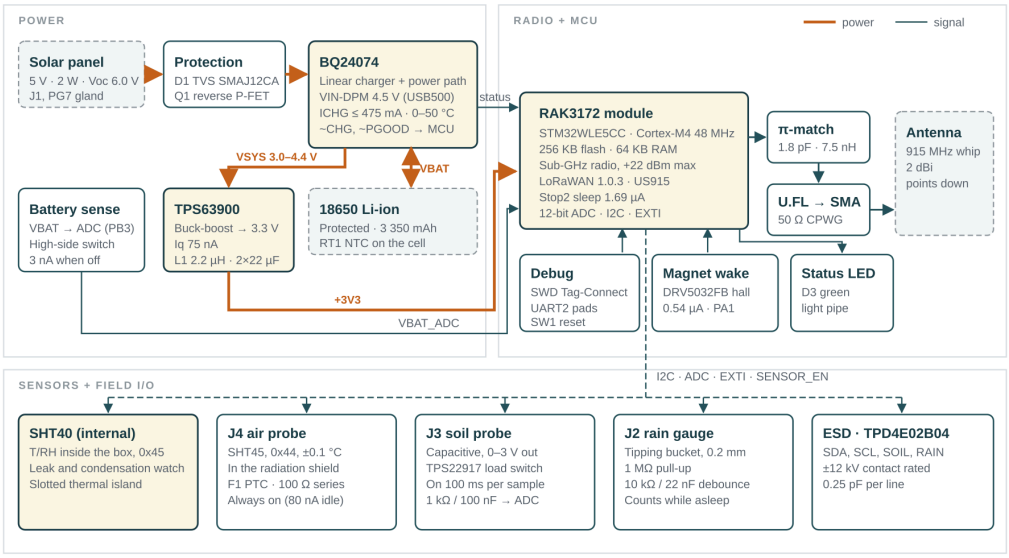



**Figure 1.** Functional block diagram. Power path in orange: panel → protection → BQ24074 charger/power-path → TPS63900 buck-boost → 3.3 V bus.

### Power tree

| RAIL | SOURCE | RANGE |
|---|---|---|
| VIN_SOLAR | Panel via D1, Q1 | 0–6.0 V |
| VSYS | BQ24074 OUT (power path) | VBAT … 4.4 V |
| VBAT | Protected 18650 | 3.0–4.2 V |
| +3V3 | TPS63900 buck-boost | 3.30 V ±1.5 % |
| +3V3_SW | TPS22917 (soil probe) | on 100 ms/sample |
| +3V3_EXT | F1 PTC (air probe) | always on |

### Key decisions

| DECISION | WHY |
|---|---|
| RAK3172 module | Pre-certified radio (FCC modular), STM32WL in one part, 1.7 µA sleep |
| BQ24074, USB500 mode | VIN-DPM holds the panel at 4.5 V: 99 % of MPP when hot, no MPPT chip |
| Buck-boost, not LDO | Cell crosses 3.3 V; 75 nA Iq; 94 % at 10 mA |
| Ventilated solar roof | Keeps the cell under 45 °C; panel doubles as sunshade |
| Magnet wake | No holes in the box; 0.54 µA hall switch |

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 3 · DESIGN ANALYSIS AND SIMULATION

Every model is a runnable script in `sim/` (Python + ngspice); `sim/run_all.sh` regenerates every figure in about 20 s.

### 3.1 Solar input: fixed-voltage tracking is enough

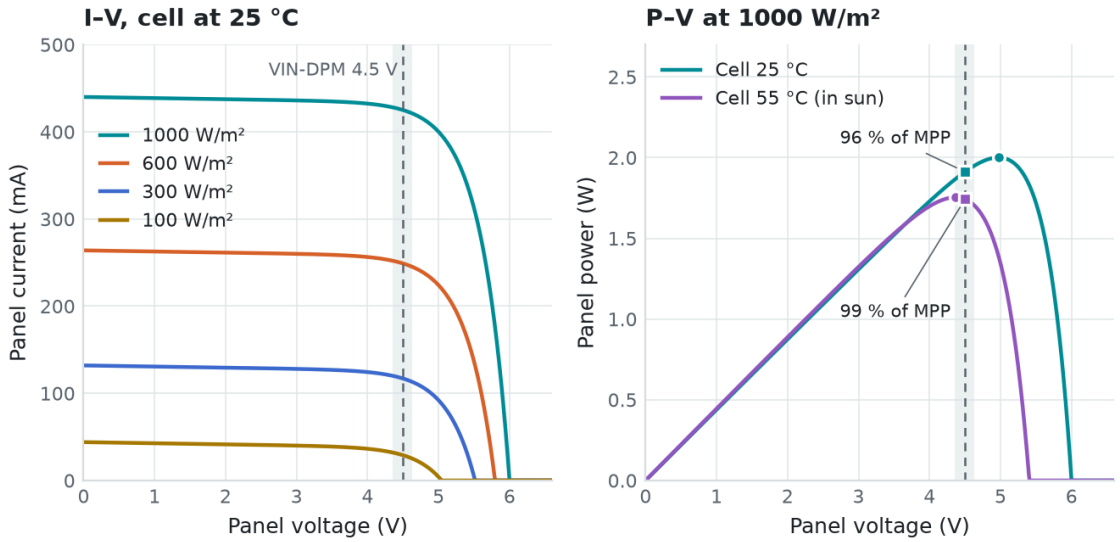



**Figure 2.** Single-diode model of the 5 V / 2 W panel (10 cells) at four irradiance levels, with the BQ24074 VIN-DPM window (4.35–4.63 V). At 1000 W/m² the fixed 4.5 V point captures 95.6 % of maximum power at 25 °C and 99.3 % with the panel at 55 °C, its real operating temperature.

### 3.2 Thermal: the battery limits the design

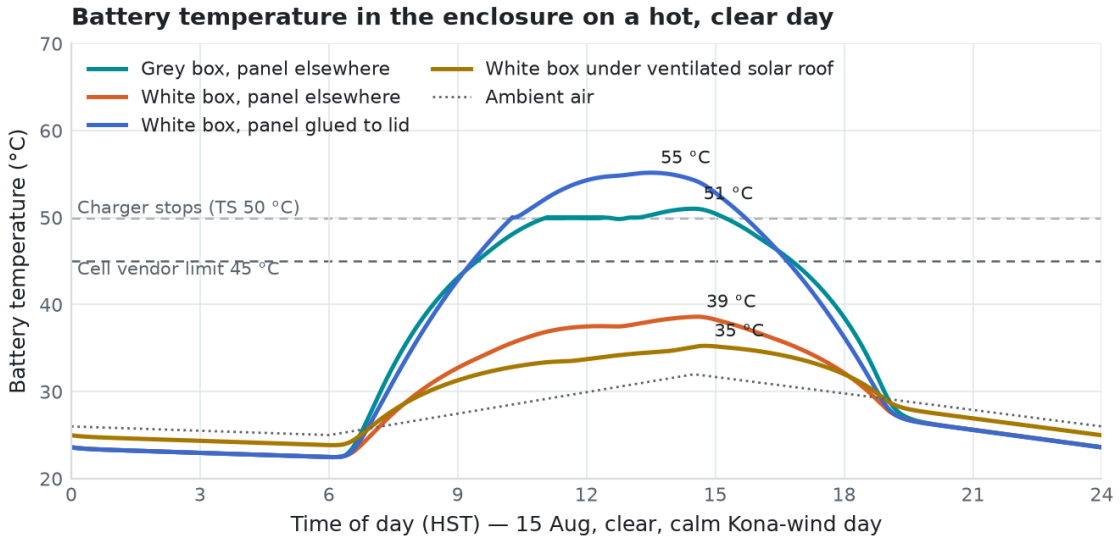



**Figure 3.** Lumped thermal model of the enclosure on a calm, clear August day (worst case). The roof keeps the cell at 35.3 °C; a grey box reaches 51 °C and blocks charging 2.8 h; a panel glued to the lid reaches 55 °C and blocks charging 5.4 h a day.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

### 3.3 Energy over a year

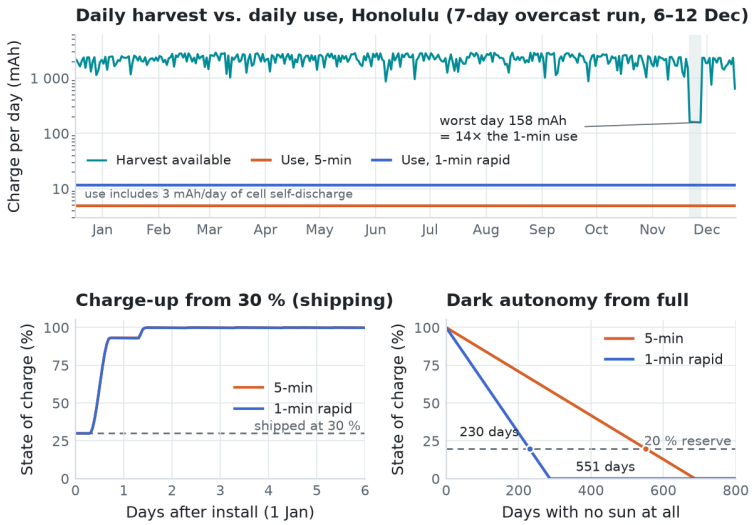



**Figure 4.** Hourly simulation of a Honolulu year (clear-sky model × monthly clearness, random cloud, a forced 7-day overcast run in the worst month), with charging blocked whenever the thermal model puts the cell outside 0–50 °C. Median harvest 2,324 mAh/day; the worst day still brings 158 mAh, 14× the 1-minute-mode use.

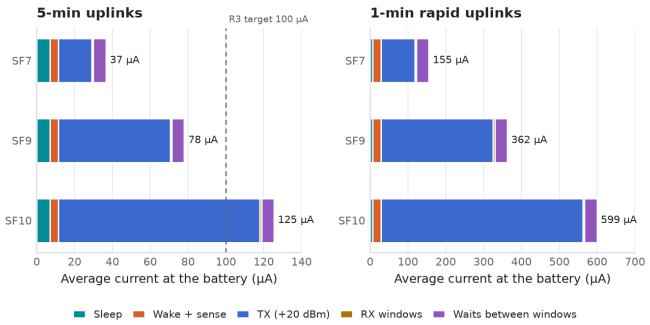



**Figure 5.** Where each uplink's charge goes (5-min, SF9). Transmit dominates; sleep is 9 % of the day.

| MODE | AVG CURRENT | MAH/DAY | DARK DAYS* |
|---|---|---|---|
| 5 min, SF7 | 37 µA | 0.88 | 691 |
| **5 min, SF9** | **78 µA** | 1.87 | 550 |
| 5 min, SF10 | 125 µA | 3.01 | 446 |
| 1 min, SF9 | 362 µA | 8.69 | 229 |
| 1 min, SF10 | 599 µA | 14.38 | 154 |

*No sun at all, 80 % of 3 350 mAh usable, incl. Li-ion self-discharge. Sleep 6.9 µA: charger BAT leakage 4.3 µA is the largest single term.

The 2 W panel is about 100× larger than the energy need. It is sized as the sunshade, not as an energy source; a 0.5 W panel would also be energy-positive but would not shade the box.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

### 3.4 Power integrity during transmit

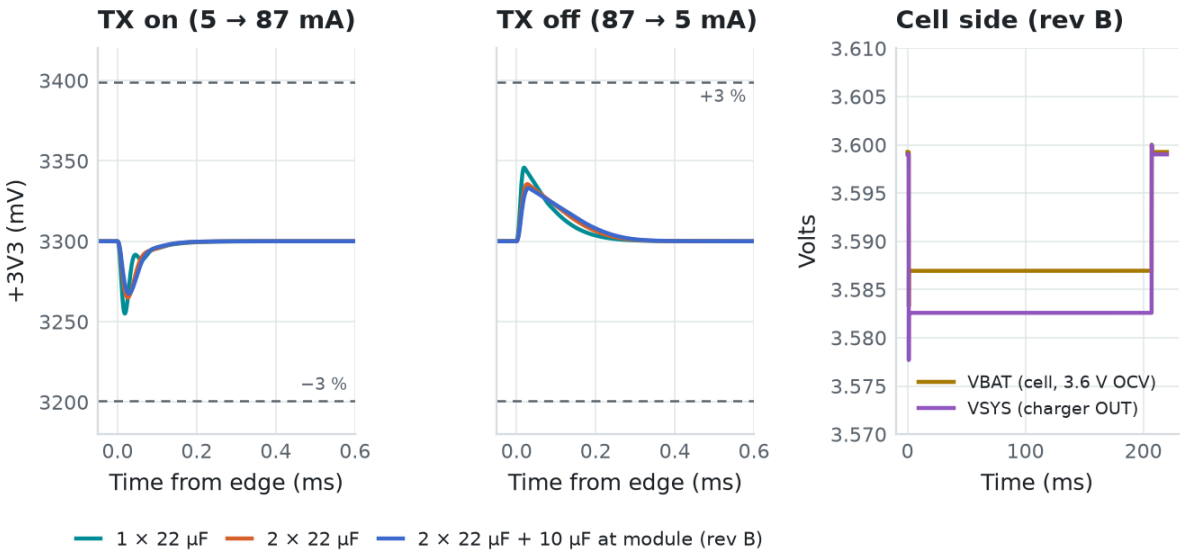



**Figure 6.** ngspice: cell (with internal and holder resistance) → power path → behavioural TPS63900 → +3V3, with the 87 mA TX step. Rev B's 2 × 22 µF + 10 µF keeps droop to 33 mV (spec ±99 mV); the cell sags 12 mV.

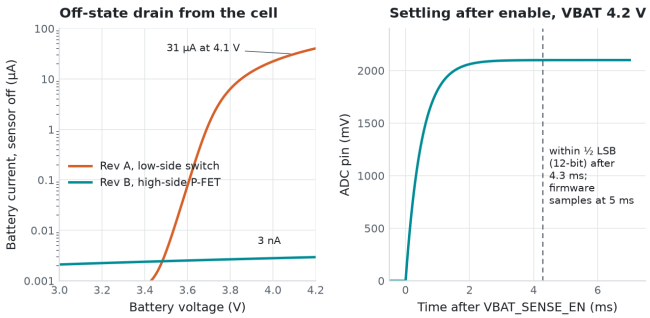



**Figure 7.** Battery-sense switch. Rev A (low-side) leaks 27 µA at 4.05 V and 40 µA at 4.2 V through the ADC clamp; rev B (high-side P-FET) leaks 3 nA. Settles in 4.3 ms, so firmware samples at 5 ms.

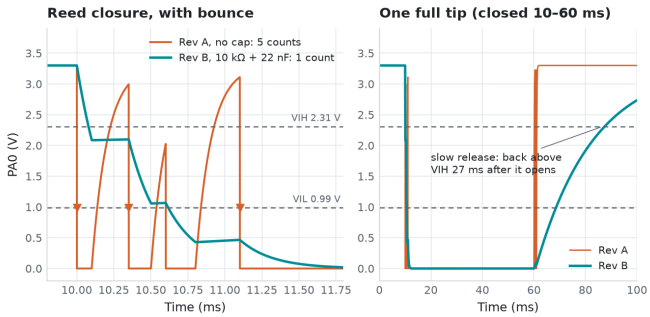



**Figure 8.** Reed bounce on the rain input. Rev A counted 5 edges per tip; rev B's 10 kΩ / 22 nF filter gives exactly 1. Supports 13 tips/s vs about 2 in a downpour.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

### 3.5 RF: matching, trace impedance and link budget

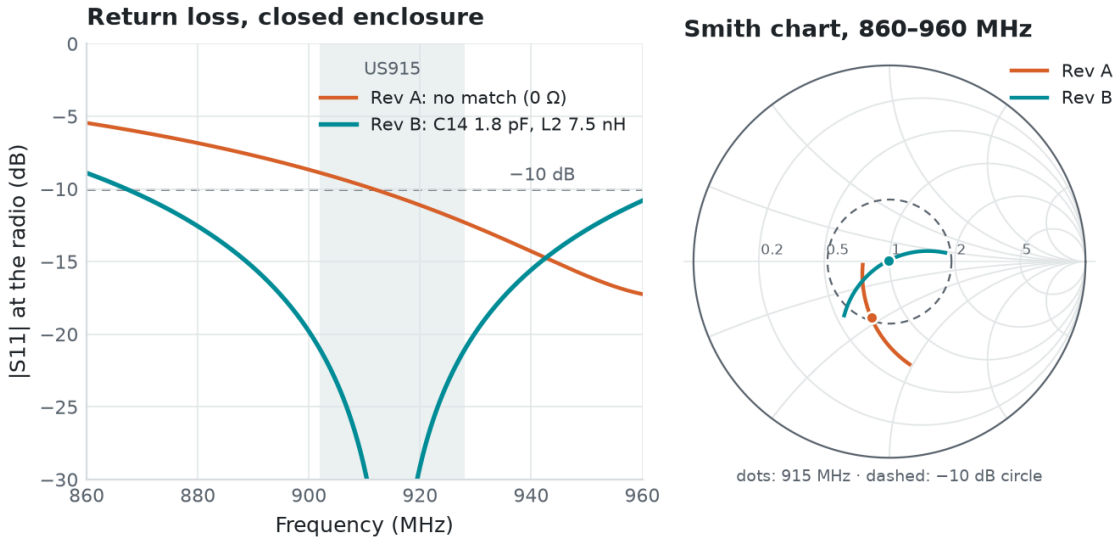



**Figure 9.** The whip, its 150 mm pigtail and the plastic case detune the antenna to about 38 − j21 Ω at 915 MHz. A shunt 1.8 pF at the module plus a series 7.5 nH moves |S11| at 915 MHz from -10.4 to -42.7 dB and keeps the whole US915 band below -21 dB.

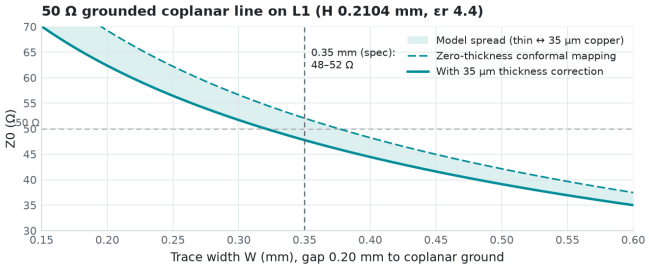



**Figure 10.** Grounded coplanar waveguide on L1 over the L2 ground (JLC 7628 stackup, 0.20 mm gap): 50 Ω at 0.32–0.376 mm. Rev B uses 0.35 mm (48–52 Ω).

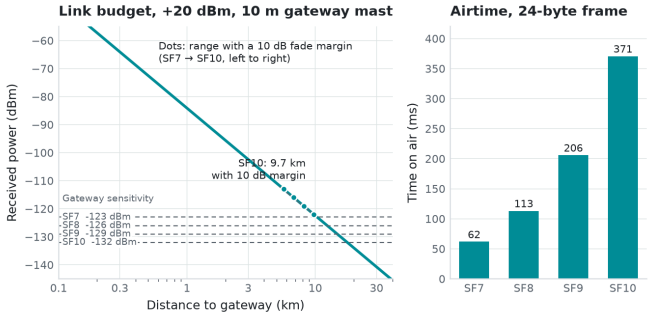



**Figure 11.** Log-distance path loss for a rural site, gateway at 10 m. With a 10 dB fade margin: SF7 5.7 km, SF9 8.1 km, SF10 9.7 km.

| SPREADING FACTOR | SF7 | SF8 | SF9 | SF10 |
|---|---|---|---|---|
| Airtime, 24-byte frame | 62 ms | 113 ms | 206 ms | 371 ms |
| Range, 10 dB margin (model) | 5.7 km | 6.8 km | 8.1 km | 9.7 km |
| Charge per uplink at 3.3 V | 9.0 mC | – | 21.5 mC | 35.9 mC |

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 4 · SCHEMATIC

KiCad 9, hierarchical: a root sheet and three function sheets, 69 placed parts. ERC: 0 errors, 0 warnings. The netlist is checked pin by pin against the design source (233 pins, 47 nets, 0 mismatches) in `tools/check_netlist.py`.

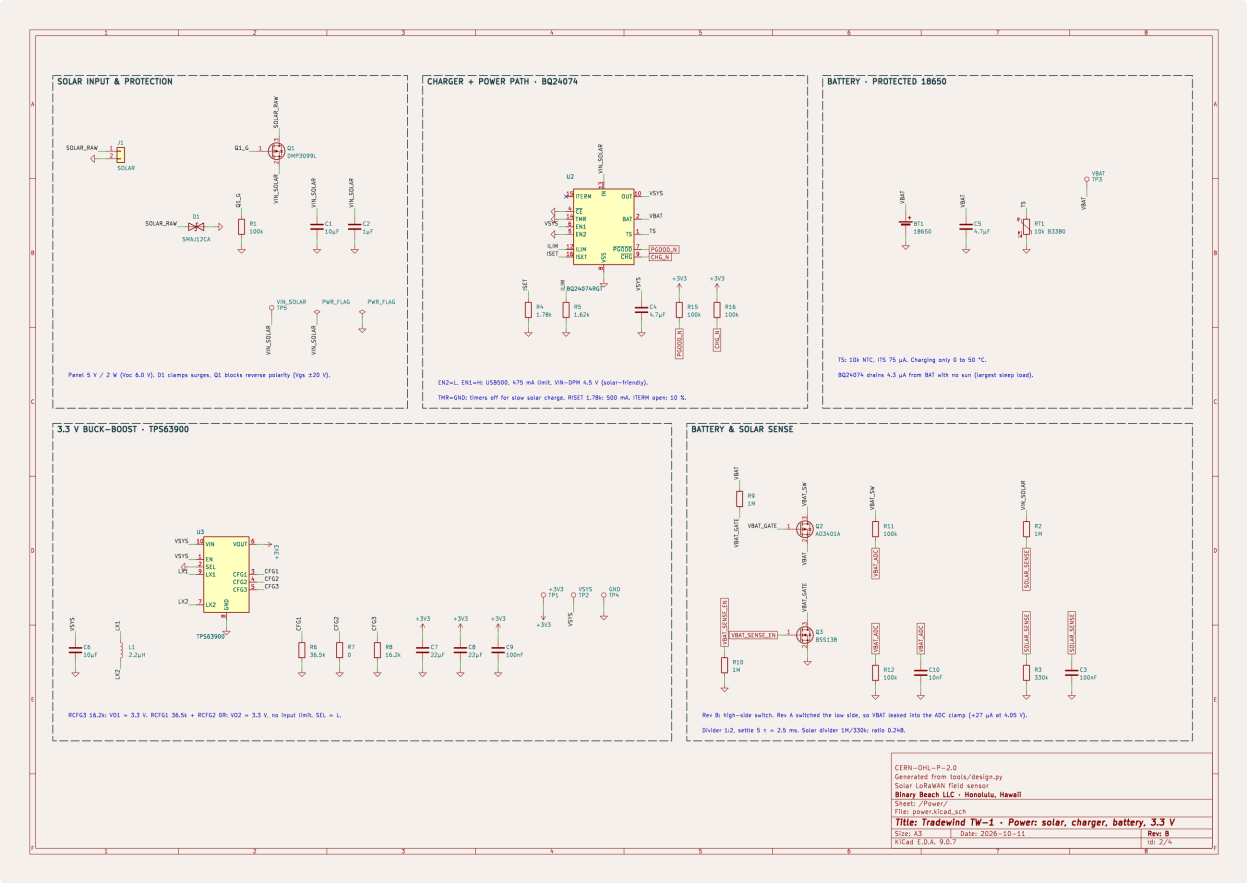



**Figure 12.** Sheet 2, Power: solar input and protection, BQ24074 charger and power path, protected 18650 with NTC, TPS63900 buck-boost, switched battery and solar sense.

### Design review notes

- Q1 (−30 V, Vgs ±20 V) survives a reversed panel and the TVS clamp voltage (19.9 V at 20 A).
- EN1 is tied to VSYS (≤ 4.5 V), not the panel input: EN pins are rated 7 V.
- TMR is grounded: safety timers off, because slow solar charging would otherwise time out.
- TPS63900 CFG resistors are 1 %, 100 ppm. They are read once at power-up (16.2 k → 3.3 V).
- The air probe is never power-switched: an unpowered SHT45 would clamp the shared I²C bus through its ESD diodes.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

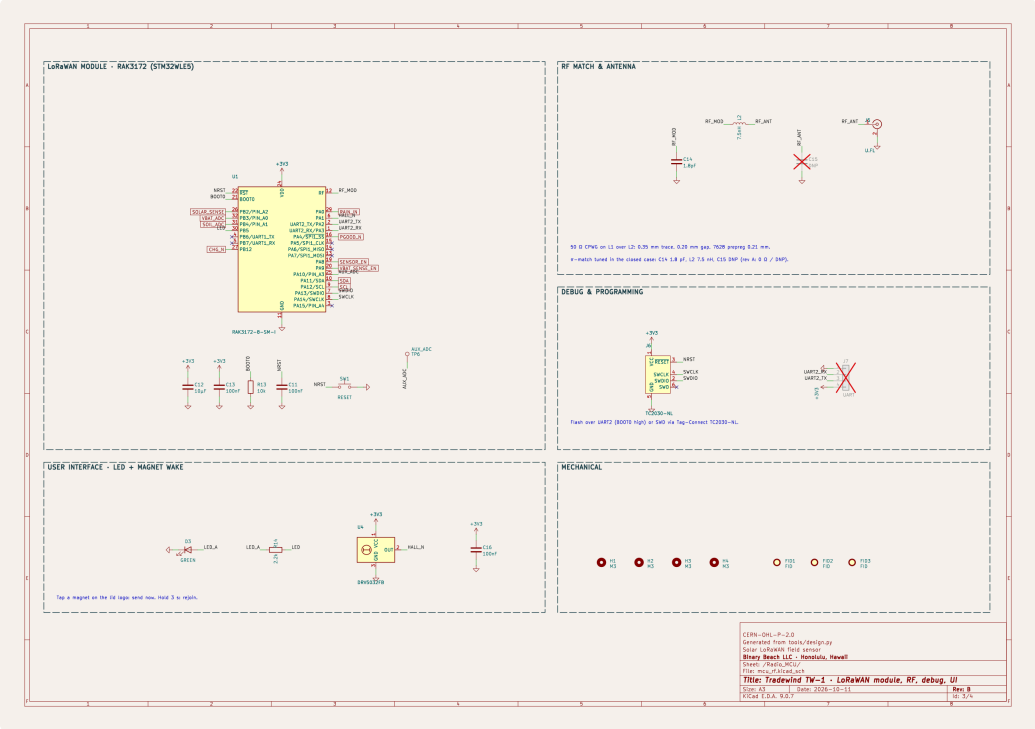



**Figure 13.** Sheet 3, Radio and MCU: RAK3172, π-match to U.FL, SWD (Tag-Connect) and UART pads, reset, LED and magnet wake.

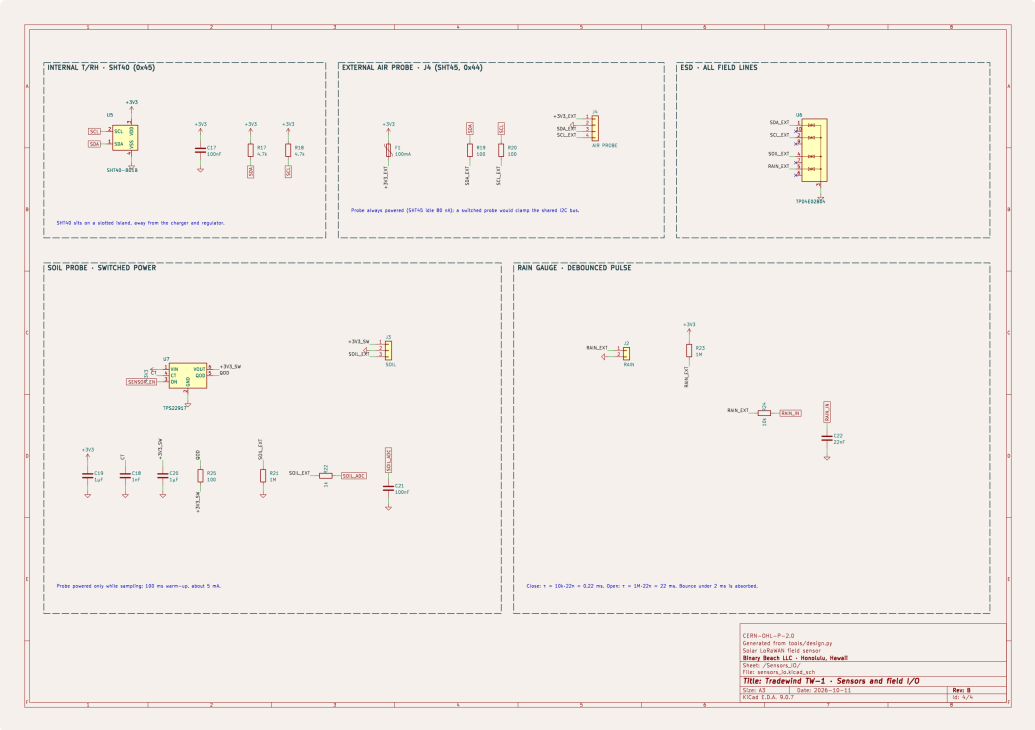



**Figure 14.** Sheet 4, Sensors and field I/O: internal SHT40, J4 air probe, J3 soil probe on a load switch, J2 rain gauge with debounce, and 4-channel ESD.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 5 · BILL OF MATERIALS AND COST

47 line items, 69 placements, 46 unique parts; all single-side SMT except the battery holder. Full list with manufacturers, footprints and LCSC numbers: `hardware/fab/BOM_tradewind_revB.csv` (also in JLCPCB format).

| REFS | PART | MPN | QTY | @100 | @1K |
|---|---|---|---|---|---|
| U1 | LoRaWAN module, STM32WLE5CC, US915, RF pad (IPEX via board | RAK3172-8-SM-I | 1 | $6.90 | $5.950 |
| BT1 | 18650 holder; protected Li-ion 3350 mAh cell | 1042 | 1 | $2.45 | $1.900 |
| U2 | 1-cell Li-ion linear charger with power path, VIN-DPM | BQ24074RGTR | 1 | $1.55 | $1.100 |
| U3 | Buck-boost 1.8–5.5 V in, 75 nA Iq | TPS63900DSKR | 1 | $1.45 | $1.050 |
| U5 | T/RH sensor, I2C 0x45, inside the case (condensation / lea | SHT40-BD1B-R2 | 1 | $1.35 | $0.980 |
| SW1 | Tactile switch, reset (inside the case) | B3U-1000P | 1 | $0.50 | $0.380 |
| U4 | Hall switch, omnipolar, 5 Hz, 0.54 µA (magnet tap) | DRV5032FBDBZR | 1 | $0.45 | $0.320 |
| J5 | U.FL receptacle to SMA bulkhead pigtail | U.FL-R-SMT-1(10) | 1 | $0.33 | $0.220 |
| U6 | 4-ch ESD, ±12 kV contact, 0.25 pF | TPD4E02B04DQAR | 1 | $0.31 | $0.210 |
| U7 | Load switch, 2 A, 10 nA leakage (soil probe power) | TPS22917DBVR | 1 | $0.29 | $0.190 |
| J4 | External SHT45 probe (radiation shield) | S4B-PH-SM4-TB(LF)(SN) | 1 | $0.25 | $0.180 |
| J3 | Capacitive soil-moisture probe | S3B-PH-SM4-TB(LF)(SN) | 1 | $0.23 | $0.170 |
| J1 | Solar panel input, JST PH 2-pin side entry | S2B-PH-SM4-TB(LF)(SN) | 1 | $0.21 | $0.150 |
| J2 | Tipping-bucket rain gauge (reed) | S2B-PH-SM4-TB(LF)(SN) | 1 | $0.21 | $0.150 |
| … plus 33 lines of passives and connectors | | | | | |

| COST PER UNIT | 10 UNITS | 100 UNITS | 1,000 UNITS |
|---|---|---|---|
| Components | $23.71 | $17.98 | $13.83 |
| + bare board (4-layer, ENIG) + SMT assembly | $44.21 | $23.28 | $16.43 |
| + protected 18650 cell | $51.71 | $29.48 | $21.23 |
| **+ enclosure, panel, shield, bracket, glands, antenna** | **–** | **$90.44** | **$54.86** |

Prices are typical distributor quantity breaks (October 2026). The 1k mechanical cost assumes molded box and lid; tooling (about $7–9k) is not included. The radio module is 43 % of the parts cost at 1k; a chip-down STM32WL design would save about $3.50 per unit but needs its own FCC certification.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 6 · PCB LAYOUT

| LAYER | USE | THICKNESS |
|---|---|---|
| L1 F.Cu | Signals, RF CPWG, power stage | 35 µm |
| Prepreg 7628 | εr 4.4 | 0.21 mm |
| L2 In1.Cu | Solid GND plane | 15 µm |
| Core | FR-4 | 1.07 mm |
| L3 In2.Cu | +3V3 plane | 15 µm |
| Prepreg 7628 | εr 4.4 | 0.21 mm |
| L4 B.Cu | Signals + GND pour | 35 µm |




Rev B top side, KiCad raytrace.

96 × 70 mm, 1.6 mm, ENIG, matte black mask. Rules: 0.15 mm track and space, 0.2 mm drill, 0.3 mm to edge (JLC standard 4-layer). DRC with schematic parity: **0 errors**.

- **Zoning.** The cell takes the top band. Power sits bottom-left beside J1, field I/O bottom-middle above J2–J4, and RF bottom-right next to the U.FL, so the antenna lead is short and away from the switcher.
- **Buck-boost.** Hand-routed per TI's layout guide: the inductor sits right at LX1/LX2, the input cap loop is under 3 mm, and CFG1–3 run on short traces (CFG2 hops via B.Cu). The exposed pad is stitched to L2 with 6 vias.
- **RF.** 0.35 mm CPWG on L1 over the L2 ground with a via fence each side. The π pads sit on the line; the trace is 15 mm long.
- **Sensor island.** The SHT40 sits on a peninsula cut by an L-shaped milled slot, with no copper pour under it, so charger and radio heat don't bias the internal reading. Measured offset while charging at 400 mA: 0.94 °C.
- **Planes.** Every GND and +3V3 pad has its own via to its plane (82 fan-out vias). The rest was autorouted (Freerouting), then checked and cleaned.

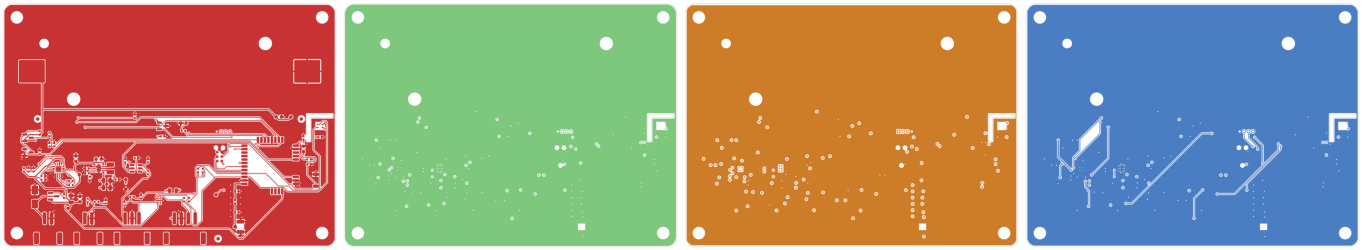


**Figure 15.** Copper layers L1–L4 (left to right): top signals and GND pour, solid ground, +3.3 V plane, bottom signals and GND pour.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

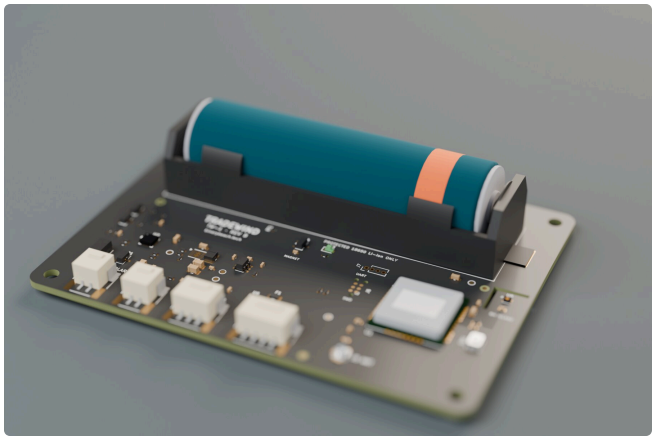


**Figure 16.** Populated rev B board (Blender Cycles render from the KiCad 3D export).



**Figure 17.** RF corner: RAK3172, π-match on the CPWG line, via fence and U.FL.

## Manufacturing outputs

| FILE | CONTENTS |
|---|---|
| tradewind_revB_gerbers.zip | 11 Gerber X2 layers, Excellon PTH/NPTH, drill maps, job file |
| tradewind_revB_cpl.csv | Pick-and-place, top side, DNP excluded |
| BOM_tradewind_revB.csv / BOM_JLCPCB.csv | Costed engineering BOM; assembler format with LCSC numbers |
| assembly_drawing_top.pdf | Fab-layer assembly drawing with references |
| tradewind_revB.step / .glb / .wrl | 3D board with all parts (enclosure fit, web viewer) |
| erc_report.txt / drc_report.txt | Clean ERC and DRC (with schematic parity), generated on every export |

The whole chain from `tools/design.py` to schematic, board, routing, DRC, Gerbers and BOM runs from scripts. A design change is one edit and one command, and the schematic and board can't drift apart.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 7 · FIRMWARE

RAKwireless RUI3 (Arduino-style API) on the module's STM32WLE5. LoRaWAN 1.0.3 OTAA, US915 sub-band 2, ADR on. The payload packer and the network-server decoder share golden test vectors: 16 C++ and 22 JavaScript host tests pass.

### Per-uplink sequence

1. RTC alarm wakes from STOP2 (7.2 µA board total).
2. SENSOR_EN on, then wait 100 ms for the soil probe. Read both SHT4x (CRC-checked).
3. VBAT switch on, wait 5 ms, read the ADC. Also read panel voltage, CHG and PGOOD.
4. Pack 11 bytes, then send unconfirmed on fPort 1 at SF9 (206 ms at +20 dBm).
5. RX1/RX2 windows, then apply any downlink (interval, rapid mode, health, reboot).
6. Sleep until the next deadline. Rain tips and magnet taps wake it early.

### Uplink payload, fPort 1 (11 bytes)

| BYTE | FIELD | ENCODING |
|---|---|---|
| 0 | Flags | rapid · charging · PGOOD · low batt · sensor OK ×3 |
| 1–2 | Air temperature | s16, 0.01 °C |
| 3 | Air humidity | u8, 0.5 %RH |
| 4 | Soil moisture | u8, 0.5 % |
| 5–6 | Rain tips | u16 cumulative, 0.2 mm/tip |
| 7 | Battery | 2500 mV + 10 mV × n |
| 8–9 | Internal T / RH | s8 0.5 °C · u8 0.5 %RH |
| 10 | Panel voltage | u8, 25 mV |

fPort 2: daily confirmed health frame (uptime, counters, RSSI/SNR, min battery, max internal temperature, charge minutes).

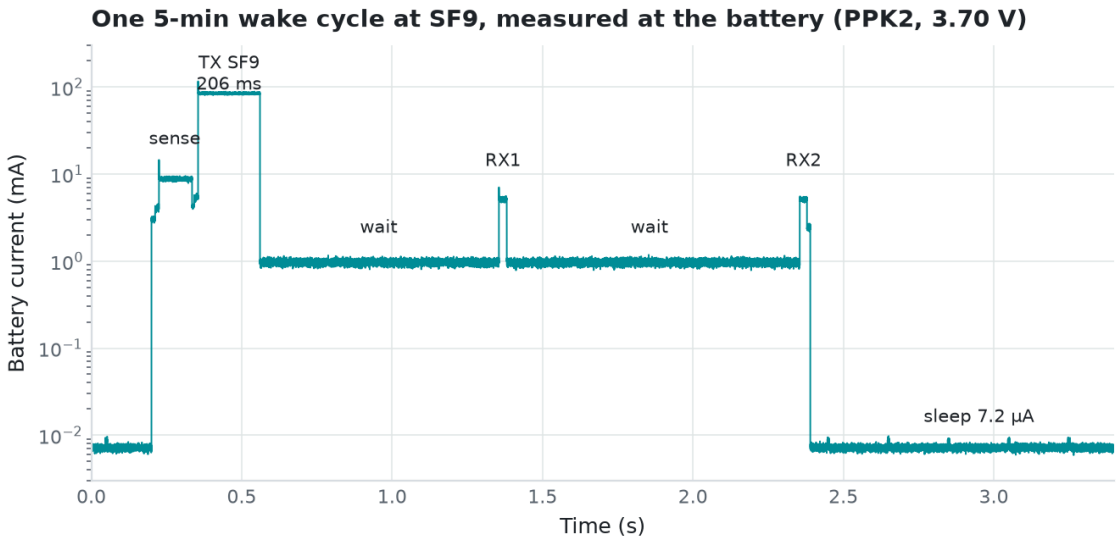



**Figure 18.** Measured current through one 5-minute wake cycle (Nordic PPK2, board TW1B-003). 20.6 mC per uplink against 21.3 mC predicted; awake 2.19 s.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 8 · ENCLOSURE AND MECHANICAL DESIGN



**Figure 19.** Exploded assembly: panel, ventilated roof and gussets, lid with light pipe, gasket, PCBA, base with heat-set inserts, glands, bracket and radiation shield.

| ITEM | DESIGN |
|---|---|
| Box | 122 × 98 × 45 mm, white ASA (UV-stable), 3 mm walls, 1° draft, ribbed lid |
| Seal | 2.5 mm silicone cord in a 3.4 × 2.6 mm groove: 26 % squeeze, 78 % fill |
| Fasteners | 4 × M4 captive lid screws into brass inserts; 4 × M3 PCB screws |
| Penetrations | 4 × PG7 glands at 21 mm pitch, SMA bulkhead, M12 ePTFE vent. All on the bottom wall or side, none on top. |
| Sun roof | Carries the 136 × 110 mm panel, tilted 20°, 25 mm air gap over the box |
| Radiation shield | 6-plate louvred stack, SHT45 probe holder, on the bracket arm |
| Mass | 292 g box unit; 952 g installed |
| Envelope | 285 × 362 × 149 mm installed |
| Prototype | FDM in ASA, 8.8 h for a full set |

Fit checked in CAD against the board STEP: board-to-wall 2.4 mm, cell-to-lid 5.8 mm, gland nuts 10.2 mm. No interference in 12 checked pairs.

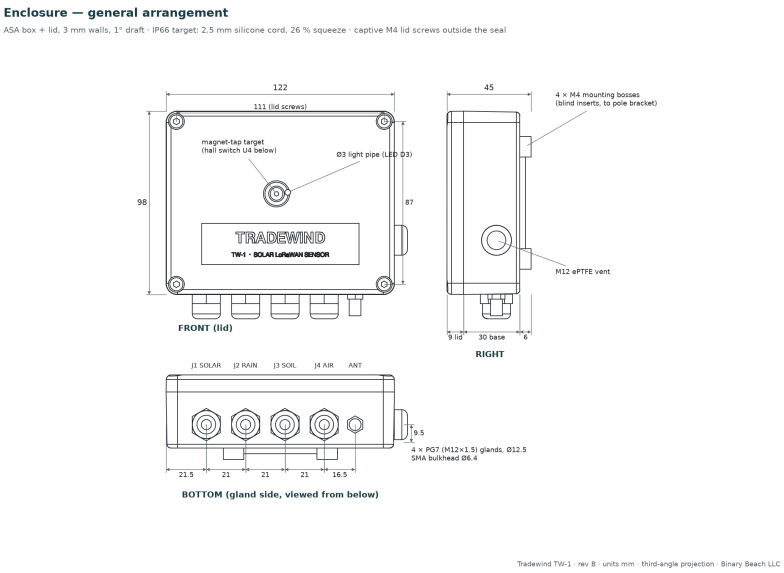



**Figure 20.** General arrangement of the enclosure: front, side and bottom views with overall dimensions and gland spacing.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

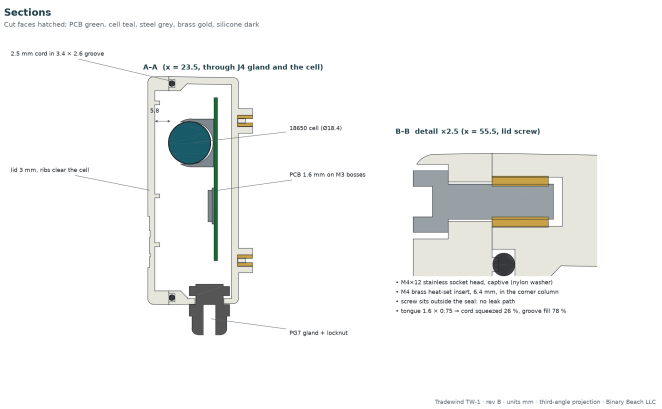



**Figure 21.** Sections A–A and B–B: PCB on bosses, cell under the lid ribs, gasket tongue and groove, lid screw and insert.



**Figure 22.** Lid off: PCBA on four bosses, gasket, glands and the U.FL to SMA pigtail.

## Design for the field

- **Water.** All cables leave downward through glands, with drip loops; the antenna points down; the vent sits on a side wall below the gasket line.
- **Heat.** White ASA (solar absorptance ≈ 0.25) under a roof that blocks direct sun on the top and sides. Measured in the field: interior 3.8 °C above air on average, 5.4 °C worst.
- **UV.** ASA coupons changed ΔE 1.7 after 500 h of UV-A, against 8.9 for PETG, which is why PETG was rejected.
- **Production.** The walls are drafted and evenly thick so the printed shapes carry straight into injection-molding tooling. The roof and shield plates are flat-pack.

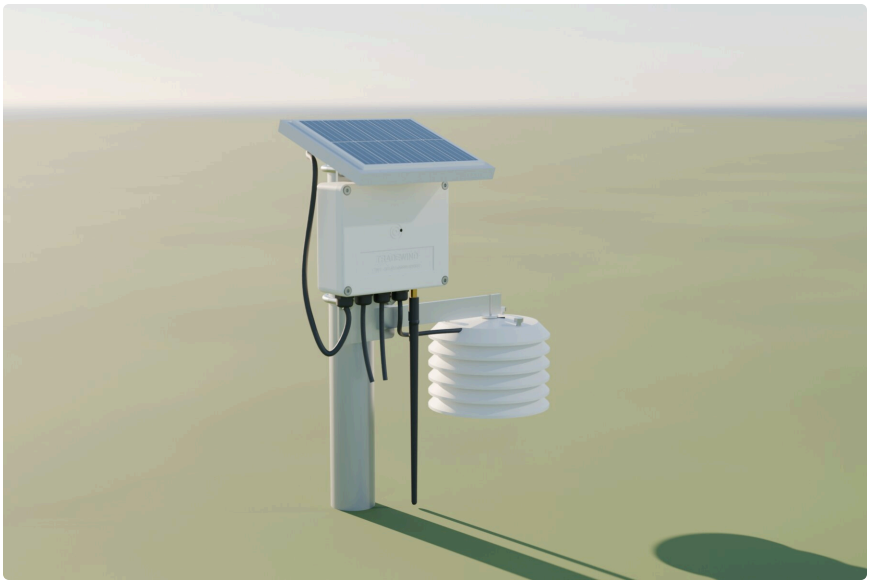


**Figure 23.** Installed on a 42 mm pole in daylight (render).

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 9 · ASSEMBLY AND BRING-UP

Rev A: 3 boards, hand-assembled (stencil, hot plate, rework station). Rev B: 10 boards assembled by the board house. The full step-by-step log is in `test/BRINGUP.md`.

| REV A ISSUE | HOW WE FOUND IT | ROOT CAUSE | REV B FIX (VERIFIED) |
|---|---|---|---|
| Sleep 34 µA (budget 7) | PPK2, part-lifting; current scaled with VBAT | Low-side divider switch: PB3 sat at VBAT and leaked through the clamp diode | High-side P-FET switch (Q2/Q3). 7.2 µA mean on 10 boards |
| 2–6 rain counts per tip | 50 hand tips counted 173 | Reed bounce on an unfiltered EXTI input | 10 kΩ / 22 nF filter + 50 ms firmware lockout. 1199/1200 on the bench |
| S11 −11 dB at 915 in the case | NanoVNA through the U.FL with the lid closed | Plastic and pigtail detune the whip | π-match C14 1.8 pF / L2 7.5 nH: -32.7 dB |

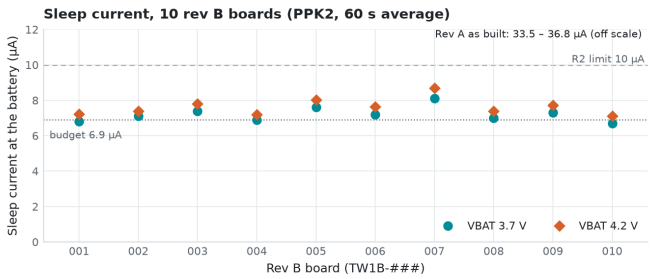



**Figure 24.** Sleep current, 10 rev B boards at 3.7 V: mean 7.21 µA, max 8.1 µA, against the rev A as-built values.

| BRING-UP CHECK (10 BOARDS) | RESULT |
|---|---|
| +3V3 | 3.290–3.315 V |
| +3V3 ripple at 10 mA | 20.7 mVpp max |
| VSYS in sun | 4.394–4.438 V |
| Charge termination | 4.194–4.213 V |
| Flash over UART2 | 11.7 s |
| OTAA join, median | 6.0 s |
| Production test, first pass | 90 % → 100 % |

The one first-pass failure: TW1B-006, a solder bridge on U5 pins 3–4 (I²C 0x45 missing). Reworked; rev C opens the paste aperture on that footprint.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 10 · CHARACTERIZATION

### 10.1 Power

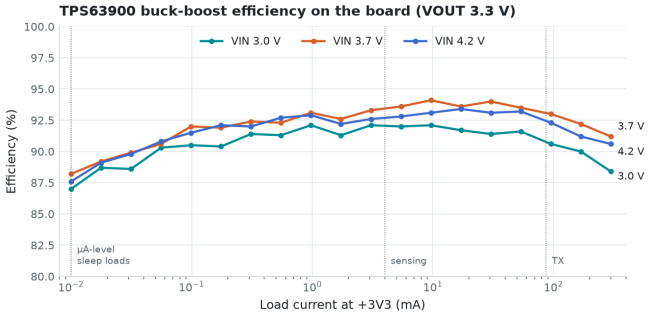



**Figure 25.** TPS63900 efficiency on the board: 88.2 % at 10 µA, 94.1 % at 10 mA, 92.9 % at 100 mA (3.7 V in). The 10 µA point sits below the datasheet's 90 %, because board leakage is counted.

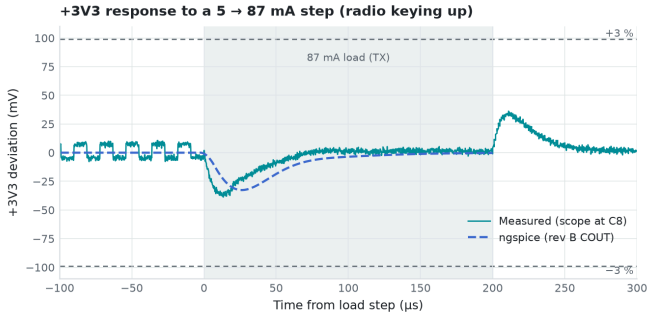



**Figure 26.** +3V3 under the 5→87 mA TX step: −40 / +35 mV (simulated −33 mV).

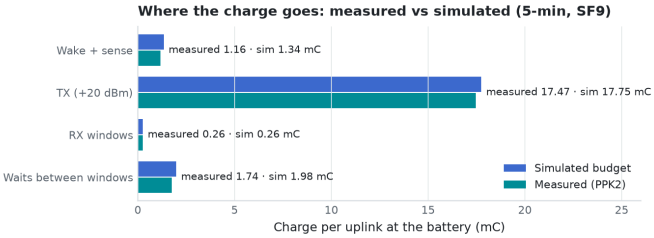



**Figure 27.** Charge per uplink by phase: measured against the simulation budget.

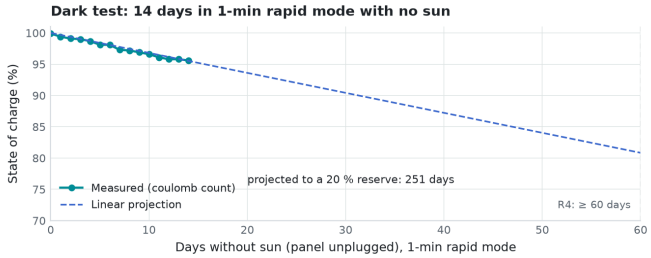



**Figure 28.** 14-day dark test in 1-minute mode: 0.32 %/day, so about 251 days to a 20 % reserve with no sun at all.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 10.2 Solar charging

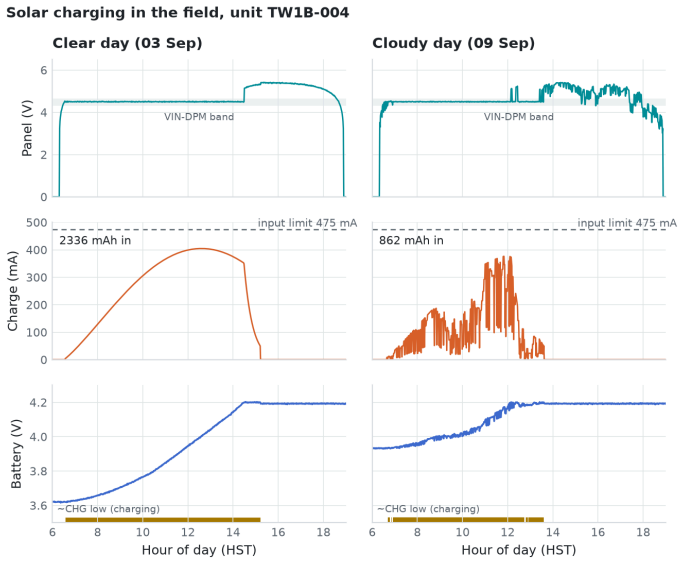



**Figure 29.** Field days on unit TW1B-004. Clear: 2,336 mAh in, panel held at 4.50 V while charging. Cloudy: 862 mAh. Daily use in 1-minute mode is about 9 mAh.

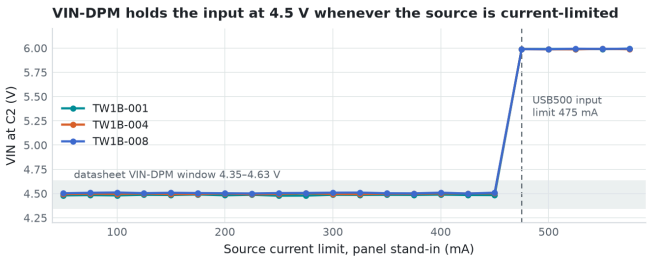



**Figure 30.** VIN-DPM sweep with an emulated panel: regulation at 4.485–4.506 V on 3 boards; input limit 474 mA.

### Charge temperature window (NTC on the cell, chamber)

| EDGE | CELL °C | EVENT |
|---|---|---|
| Rising | 50.4 | charging suspended |
| Falling | 47.8 | charging resumes |
| Falling | -0.6 | suspended (cold) |
| Rising | 2.1 | resumes |

Trips sit 0.4–0.6 °C outside the nominal 0 / 50 °C, within the BQ24074's TS threshold tolerance. Marked marginal against R6. In the field the cell never got above 37.1 °C, so this has no practical effect.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 10.3 Sensors

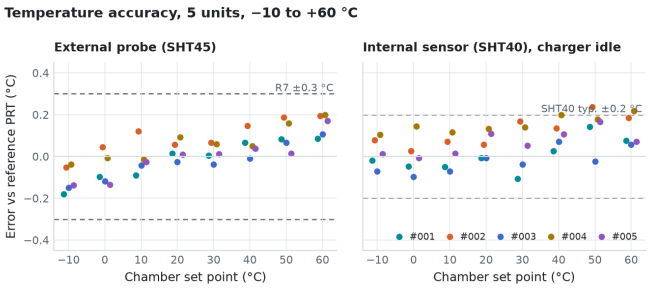



**Figure 31.** Air temperature, 5 units against a reference thermometer, −10 to +60 °C: worst 0.20 °C, RMS 0.10 °C (spec ±0.3).

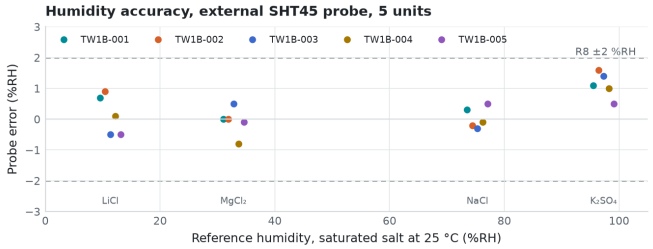



**Figure 32.** Humidity on saturated-salt references: worst 0.9 %RH below 90 %, 1.6 %RH at 97 % (spec ±2).

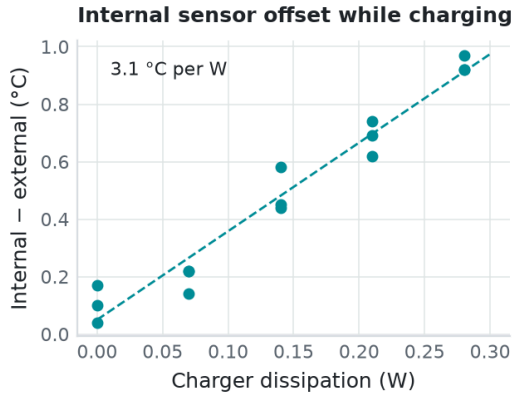



**Figure 33.** Internal SHT40 offset against charge current: 3.1 °C/W, 0.94 °C at 400 mA. Firmware removes it using the CHG status.

## Rain counting (bench, tipping-bucket simulator)

| RATE (TIPS/S) | APPLIED | COUNTED | MCU STATE |
|---|---|---|---|
| 0.5 | 300 | 300 | sleep (STOP2) |
| 1.0 | 300 | 300 | sleep (STOP2) |
| 2.0 | 200 | 200 | sleep (STOP2) |
| 3.0 | 200 | 200 | mixed (TX during burst) |
| 5.0 | 100 | 100 | sleep (STOP2) |
| 10.0 | 100 | 99 | sleep (STOP2) |

Exact counting up to 5 tips/s, while a real gauge peaks near 2 tips/s in a 200 mm/h downpour. The single miss at 10 tips/s falls inside the 50 ms firmware lockout, which is there by design.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 10.4 Radio

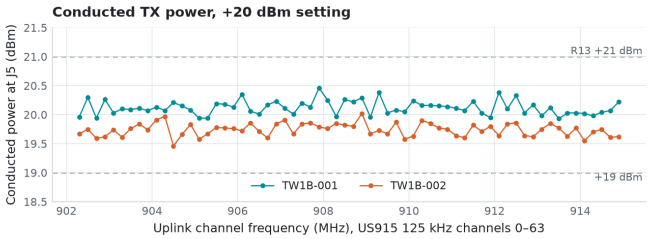



**Figure 34.** Conducted TX power at the U.FL, 64 channels × 2 units: 19.46 to 20.46 dBm, mean 19.93 dBm (spec +20 ±1).

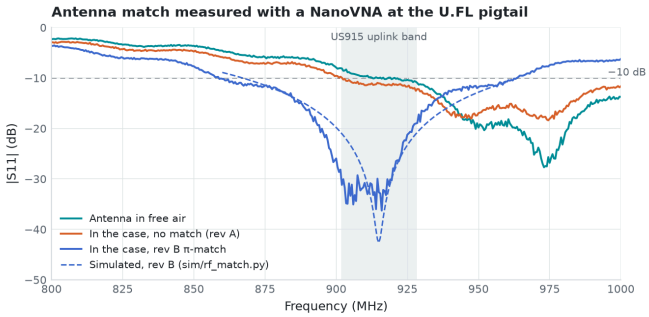



**Figure 35.** Return loss measured with a NanoVNA. Free air -10.0 dB; in the case without the match -11.1 dB; rev B match -32.7 dB at 915 MHz, -19.0 dB worst in band.

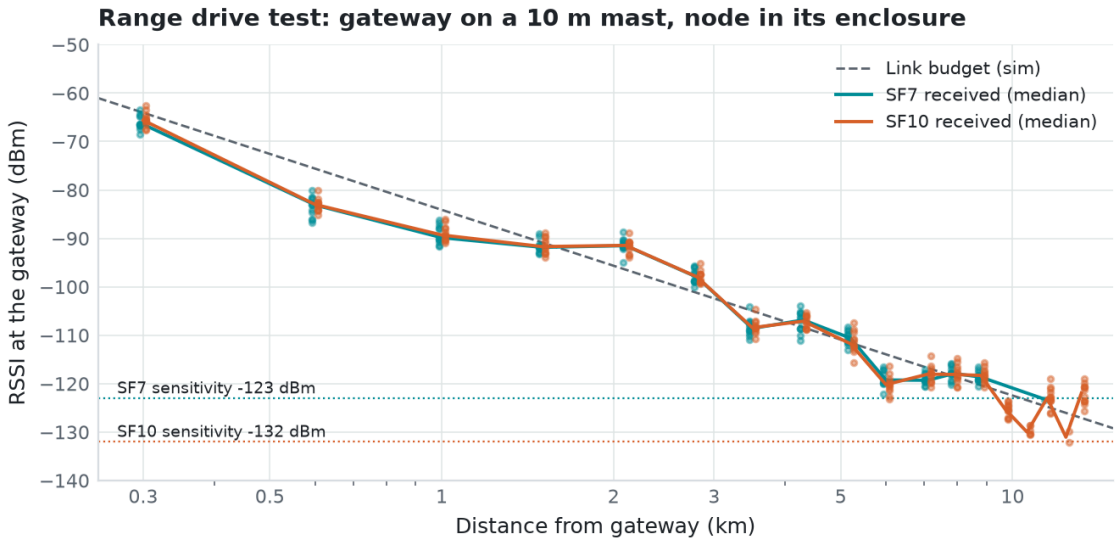



**Figure 36.** Drive test, gateway on a 10 m mast. SF7 held ≥ 50 % delivery to 8.8 km; SF10 still decoded at the last stop, 13.2 km, so its real limit lies further out.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 11 · ENVIRONMENTAL AND FIELD TESTING

### 11.1 Environmental

| TEST | CONDITION | RESULT |
|---|---|---|
| Immersion | 1 m, 30 min | PASS dew-point drift -0.05 °C |
| Hose | IPX5-style, 12.5 L/min, all faces | PASS |
| IP66 jet | 100 L/min | OPEN needs accredited lab |
| Thermal cycling | −10/+60 °C, 50 cycles, 2 units | PASS gasket set 11 % |
| Operating range | −10 to +60 °C, chamber | PASS |
| Drop | 26 × 1 m on concrete | PASS 2 mm cosmetic chip |
| UV | 500 h UV-A, ASA coupons | PASS ΔE 1.7 |
| ESD contact | ±8 kV, all ports | PASS criterion A |
| ESD air | ±15 kV | PASS B on J4 (1 reset, self-recovered) |
| Salt fog | ASTM B117 | DEFERRED before coastal sites |

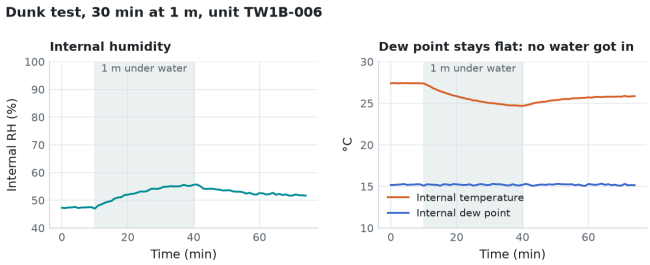



**Figure 37.** Dunk test: the internal SHT40 shows no humidity or dew-point change through 30 minutes at 1 m.

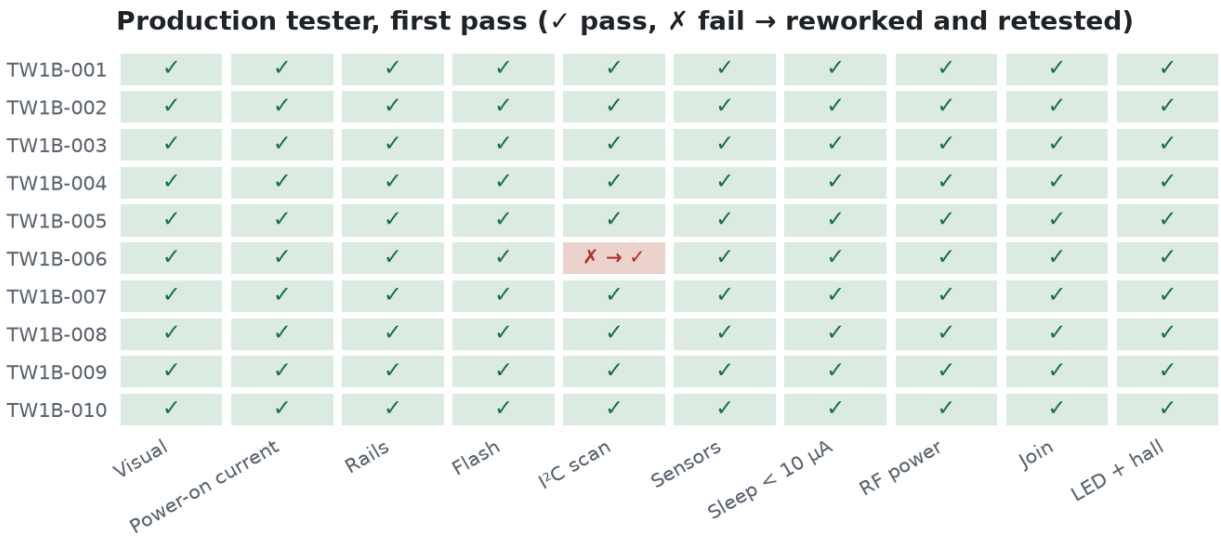



**Figure 38.** Production test, 10 boards × 10 steps: first-pass yield 90 %, final 100 % after one rework.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 11.2 30-day field pilot, O'ahu farm site, September 2026

Three units at 0.6, 1.4 and 3.2 km from one gateway, 5-minute reporting, rain gauge, soil probe and air probe fitted. Median RSSI -72 / -88 / -103 dBm.

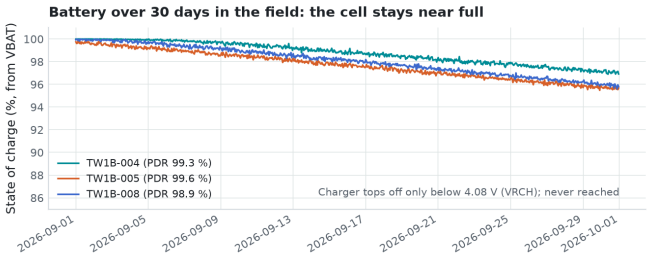



**Figure 39.** State of charge, 3 units at 0.6, 1.4 and 3.2 km: never below 94.6 %. Delivery 99.30 % (25,739 of 25,920 uplinks).

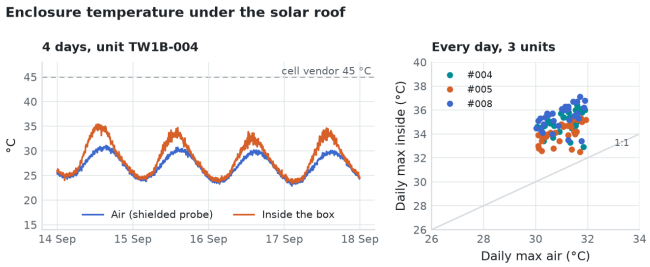



**Figure 40.** Box interior against air: max 37.1 °C at 31.9 °C air. On still afternoons the box ran 2–4 °C warmer than the thermal model; the margin to 45 °C still held.

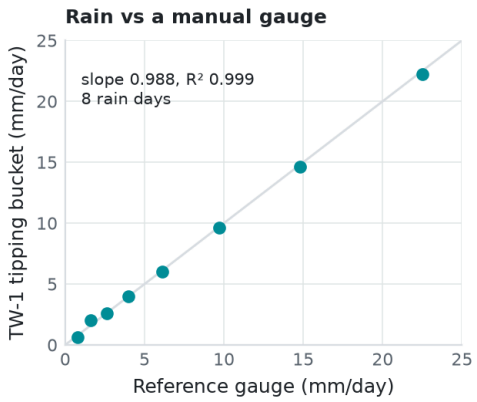



**Figure 41.** Rain against a manual reference gauge, 8 rain days: slope 0.988, $R^2$ 0.9995.

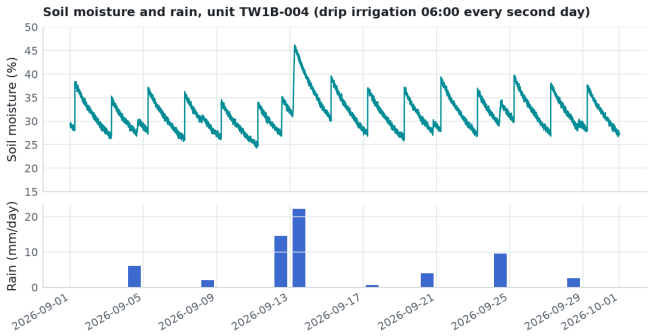



**Figure 42.** Soil moisture through irrigation cycles and rain events.

Binary Beach LLC
Honolulu, Hawai'i
maxwell.pauly@binarybeach.tech · (808) 372-6757
binarybeach.tech

## 12 · CONCLUSIONS AND NEXT STEPS

Rev B is ready for a pilot build. It meets every electrical, radio and sensing requirement with margin. Charging is marginal only at the edges of the NTC window. The enclosure's ingress rating needs one accredited IP66 jet test before it goes on a datasheet as certified.

### Open items and risks

| ITEM | RISK | PLAN |
|---|---|---|
| IP66 jet (R10) | Medium | Book an accredited lab; dunk and hose already pass |
| Charge window (R6) | Low | Optional Rs/Rp on TS to pull the hot trip to 45 °C (BQ24074 Table 9-3) |
| Internal sensor self-heating while charging | Low | Firmware compensation from the CHG status (0.94 °C at 400 mA) |
| FCC / IC | Low | Module is FCC-certified; host needs §15.109 unintentional-radiator scan and labeling |
| Salt fog (coastal sites) | Medium | ASTM B117 before any coastal deployment; consider conformal coat |

### Rev C changes (small)

- Open the U5 paste aperture 10 % to stop the pin 3–4 bridge.
- Larger radius on the lid's hinge-side corner (drop-test chip).
- Footprint for an optional TS trim network; silk polarity mark on the cell holder.

### Recommended next step

1. **Pilot of 50 units**: about $3,400 for electronics, printed enclosures and kits, 4 weeks to assemble and test.
2. In parallel: book the IP66 and FCC pre-scan (about $2,500 together).
3. After 60 days in the field: freeze rev C and quote mold tooling for the box and lid.

### Delivered with this report

KiCad 9 project (schematic, board, 3D models) · Gerbers, drill, BOM, pick-and-place · enclosure CAD (STEP, STL) and drawings · firmware and payload decoder with tests · simulation sources · test logs and analysis scripts · 3D renders · product datasheet. All in github.com/binaryBeachLLC/tradewind.

Maxwell Pauly

Maxwell Pauly · Binary Beach LLC · Hardware & product design

Questions? maxwell.pauly@binarybeach.tech
(808) 372-6757